Xiang-Ling Lin

L224749391@email.com
xxxpatty.github.io
xxxPatty

Research Interests

Networked Systems; Network Telemetry; Network Digital Twin; Programmable Network Devices; SmartNIC/DPU Offloading; Flow-level Modeling; Scalable Monitoring Architectures; Performance Evaluation

Education

2022–2024 **M.S. in Institute of Network Engineering, Department of Computer Science**, *National Yang Ming Chiao Tung University (NYCU), GPA: 4.23/4.3*
Thesis: *Design and Implementation of LAG/LACP in ns-3*

2018–2022 **B.S. in Computer Science and Engineering**, *National Taiwan Ocean University (NTOU), GPA: 93/100; Graduated 1st in the Department of Computer Science (1/120)*
Undergraduate Project: *PSABot*

Research Experience

2025–Present **Research Assistant**, *Academia Sinica, Taiwan*
Network Digital Twin (NDTwin)
- Designed a scalable network digital twin workflow driven by real sFlow telemetry for what-if scenario simulation.
- Built a high-performance telemetry collection and analysis pipeline in **C++** for scalable, near real-time network monitoring.
- Developed modules for near real-time **per-flow rate estimation** and **per-link utilization** analysis to support applications such as energy saving and traffic engineering.
- Identified telemetry sample processing as a key scalability bottleneck, motivating future exploration of **programmable-device-assisted offload** for packet parsing, aggregation, and flow-state updates.
- Built and managed a physical testbed with about **20 switches** and **10 hosts** (including traffic generators and collectors) for NDTwin development and performance evaluation.

2022–2024 **Graduate Researcher**, *Network and System Laboratory, NYCU*
M.S. Thesis: Design and Implementation of LAG/LACP in ns-3
- Evaluated the scalability and applicability of **ns-3**, **Mininet**, **OMNeT++**, and **GNS3** for large-scale network simulation.
- Motivated by the use of **LACP** in Chunghwa Telecom's network to improve resilience, implemented **LAG** and **LACP** in **ns-3** based on **IEEE 802.1AX** and Linux Bonding, extending ns-3's support for realistic wired-network simulation.
- Designed and implemented the key components needed to support link aggregation, traffic distribution, and link failure detection.
- Developed simulation scenarios to validate negotiation, failover, recovery, and flow redistribution behaviors under dynamic link conditions.
- Conducted performance evaluation on the implementation overhead under different traffic rates and transmission modes.

2019–2022 **Undergraduate Researcher**, *NTOU*
PSABot: A Chatbot System for Stack Overflow Analysis
- Built an NLP-based chatbot for retrieving and ranking relevant Stack Overflow Q&A.
- Integrated keyword extraction and relevance scoring with weighted ranking mechanisms.
- Added live discussion matching to improve answer quality and user support.

Teaching Experience

2022 **Teaching Assistant, Introduction to Computer Networks**, *National Yang Ming Chiao Tung University (NYCU),* Professor: Shie-Yuan Wang
- Graded assignments, quizzes, and exams for an undergraduate computer networks course.
- Assisted students in completing laboratory experiments and troubleshooting implementation issues.
- Guided students through experimental procedures and explained networking-related concepts during lab sessions.
- Prepared lab materials and supported course logistics for hands-on experiments.

Publications

2026 Wang, S.-Y., Chen, W.-Y., Wang, Y., **Lin, X.-L.** *A High-Performance and Scalable sFlow Scheme with Zero Control-Plane Overhead*. *IEEE International Conference on Communications (ICC)*, 2026, pp. 1–6.

2025 Wang, S.-Y., **Lin, X.-L.**, Zou, N.-C., Tsai, M.-L., et al. *Simulating the Digital Twin of a Large and High-Speed Network: Methodology and Insights*. *IEEE/ACM NOMS 2025*.

2023 Shau, A.-C., Liang, Y.-C., Hsieh, W.-J., **Lin, X.-L.**, Ma, S.-P. *PSABot: A Chatbot System for the Analysis of Posts on Stack Overflow*. *IEEE CSEE&T 2023*.

Submitted Papers

2026 Wang, S.-Y., Chen, W.-Y., **Lin, X.-L.**, Mou, T.-L., Hsieh, C.-C., Chen, Y.-C. *Equipping A Network Digital Twin with Ultra-Precision Real-Time Flow Information*. Submitted to *IEEE/ACM Transactions on Networking*.

Technical Skills

Programming	C/C++, Python
Systems	Linux, Socket Programming, Network Stack, Concurrency, DPDK
Networking	sFlow, OpenFlow, Network Telemetry, Flow-level Modeling, Link Utilization Analysis
Programmable Network Devices	P4 architecture and programmable switch concepts; SmartNIC/DPU-based packet processing and offloading
Simulation/ Emulation	ns-3, Mininet, GNS3
Evaluation Tools	Git, tcpdump/Wireshark, iperf3

Honors

2021 3rd Place, National Information Application Service Innovation Competition (Taiwan)